SOMAIYA VIDYAVIHAR UNIVERSITY

05·05·2025 (E)

Semester:– January 2025 – April 2025 Maximum Marks: 100 Examination: ESE Examination		Duration:3 Hrs.
Programme code: 01 Programme: UG (BTech)	Class: TY	Semester: VI (SVU 2020)
Name of the Constituent College/ School K. J. Somaiya School of Engineering	Name of the department: Computers	
Course Code: 116U01C601	Name of the Course: Digital Signal and Image Processing	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	20
i)	Determine whether the discrete-time signal is periodic or non-periodic: $x(n) = cos\,(\frac{3\pi n}{4})$	5
ii)	Prove that DFT matrix is symmetric? Justify your answer.	5
iii)	What is the role of a mask (or kernel) in spatial filtering? Explain with an example.	5
iv)	Compute Walsh transform of the given image.	5

2	1	2	1
1	2	3	2
2	3	4	2
1	2	3	2

v)	What effect would set to zero the lower order 2 bits in 4BPP image planes have on the histogram of an image in general?	5
vi)	Determine whether signal is energy signal or power signal: $x(n) = \cos(\frac{\pi n}{4})$	5
vii)	Explain morphological opening and closing	5

Que. No.	Question	Max. Marks
Q2 A	Explain Homomorphic filter in detail with necessary derivation and example.	10
	OR	
Q2 A	Given the following 5×5 grayscale image, apply a 3×3 median filter to all pixels except the border pixels.	10

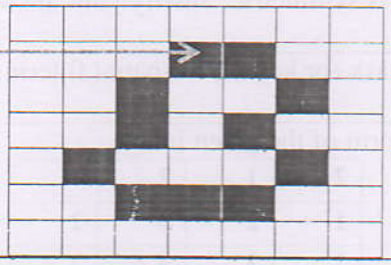
$$I = \begin{bmatrix} 10 & 200 & 10 & 200 & 10 \\ 200 & 255 & 50 & 255 & 200 \\ 10 & 50 & 0 & 50 & 10 \\ 200 & 255 & 50 & 255 & 200 \\ 10 & 200 & 10 & 200 & 10 \end{bmatrix}$$

Q 2 B	Solve	10
i)	An 8 point sequence is given by **x (n) = [1, 1, 1, 1, 0, 0, 0, 0]**. Compute 8-point DFT of x (n) by radix-2 DIT-FFT.	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	For a 3 BPP 4*4 size image , perform the following operations: 3 BPP Image a) Negation b) Thresholding with T=4 c) Intensity level slicing with background r1>=2 and r2<=5 d) Bit plane slicing for MSB and LSB planes e) Clipping with r1=2 and r2=5	10

1	2	3	0
2	4	6	7
5	2	4	3
3	2	6	1

ii)	Compute the discrete cosine transform (DCT) of the following sequence: **f(x) = {1,2,4,7}**	10
iii)	Given an image write down the 8 chain code, all the iterations have to be drawn and written down clearly. p	10



Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Describe JPEG compression in detail with necessary block diagram	10
ii)	Given four points in xy plane with the following coordinates (1,1) ,(2,2),(3,3),(4,4) use Hough transform to join these points.	10
iii)	Write down the masks for different edge detecting operators and their significance: 1) Roberts Mask 2) Prewitt Mask 3) Compass Operators 4) Sobel Mask	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any **four**	**20**
i)	Hit and Miss Transform	5
ii)	Compare and contrast point and neighbourhood spatial domain enhancement techniques.	5
iii)	Hoteling Transform	5
iv)	Vector Quantization	5
v)	Region Split and Merge based segmentation	5
vi)	Interpixel redundancy	5